

Generalized Representational Coherence: A Latent Factor Underlying LLM Trustworthiness

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Abstract

Large language models exhibit high cross-correlation ($\rho = 0.80\text{--}0.87$) across trustworthiness benchmarks, but this structure is typically dismissed as a scale confound. We propose that the correlation reflects a shared latent factor—Generalized Representational Coherence (GRC)—arising from representation stability. Through meta-analysis of 4,561 models from the Open LLM Leaderboard, we show that after controlling for $\log(\text{parameters})$ and release date, a dominant principal component persists ($\lambda_1 = 2.277$, 60% variance explained, $p = 0.001$). This residual factor correlates positively with a Behavioral Stability Index ($\rho = 0.405$, $p < 10^{-179}$) and increases systematically with instruction-tuning (Cohen’s $d = 1.87\text{--}1.99$). Prospective validation confirms generalizability: 5 of 6 holdout benchmarks load ≥ 0.3 on frozen factor weights. Our findings suggest that trustworthiness may be more unified than current multi-dimensional evaluation assumes, with implications for efficient assessment and targeted training interventions.

1 Introduction

When state-of-the-art language models achieve high scores on one trustworthiness benchmark, they systematically excel on others—even those measuring seemingly unrelated capabilities like mathematical reasoning and instruction following. Across the Open LLM Leaderboard’s 4,561 models, benchmark scores correlate at $\rho = 0.80\text{--}0.87$, a pattern too strong and too consistent to be coincidental. This observation challenges a foundational assumption in LLM evaluation: that trustworthiness dimensions like truthfulness, robustness, and reliability are independent constructs requiring separate measurement.

The standard approach treats each trustworthiness dimension as a distinct target. Specialized benchmarks—TruthfulQA for factual accuracy, MMLU for knowledge, AdvGLUE for robustness—are developed, validated, and deployed independently. Practitioners evaluate models dimension-by-dimension, constructing scorecards that imply orthogonal capabilities. Yet the high cross-correlation suggests this framing may fundamentally mischaracterize how trustworthiness manifests in language models.

We identify a deeper problem: the correlation structure itself has been treated as a nuisance confound rather than an informative signal. Prior work attributes shared variance to model scale—larger models score higher on everything—and dismisses the residual as noise. But what if the residual is the signal? What if, after controlling for scale and training recency, a dominant latent factor persists that reflects something meaningful about model behavior?

This gap—the absence of a systematic factor-analytic framework for understanding cross-benchmark correlation in LLM trustworthiness—motivates our work. We draw on psychometric methodology, where similar “positive manifold” patterns led to the discovery of general intelligence (*g*-factor), to investigate whether an analogous latent construct underlies trustworthy LLM behavior.

Our key insight is that cross-benchmark correlation reflects a shared latent factor arising from *representation stability*. Models with more stable internal representations produce consistent outputs across semantically equivalent inputs, simultaneously boosting performance on benchmarks measuring different surface-level capabilities. We term this latent factor **Generalized Representational Coherence (GRC)**.

Building on this insight, we make the following contributions:

1. **Existence of residual factor:** Through large-scale meta-analysis ($N = 4,561$ models), we demonstrate that a dominant principal component (PC1, $\lambda_1 = 2.277$, explaining 60% of residual variance) persists after controlling for $\log(\text{parameters})$ and release date, significantly exceeding permutation null thresholds ($p = 0.001$).
2. **Mechanistic evidence:** We show that PC1 correlates positively with a Behavioral Stability Index (BSI), a proxy for representation consistency ($\rho = 0.405$, $p < 10^{-179}$), supporting the interpretation that GRC reflects stability rather than general capability.
3. **Quasi-intervention evidence:** Instruction-tuning increases both BSI (Cohen’s $d = 1.87$) and PC1 scores ($d = 1.99$) within matched base/instruct model pairs, suggesting that stability-enhancing training procedures causally improve GRC.
4. **Prospective validity:** The frozen PC1 weights generalize to 5 of 6 holdout trustworthiness benchmarks (loadings ≥ 0.3), demonstrating that GRC captures a genuine latent dimension rather than benchmark-specific artifacts.

These findings suggest that evaluating trustworthiness as a single, latent factor—rather than a collection of independent dimensions—may better reflect the underlying structure of model capabilities.

2 Related Work

Our work connects three research threads: multi-dimensional trustworthiness evaluation, calibration and uncertainty quantification, and latent factor analysis

in machine learning.

2.1 Multi-Dimensional Trustworthiness Evaluation

Recent frameworks conceptualize LLM trustworthiness as a multi-faceted construct. Zhou et al. [2024] propose the Trust-RAG Compass with six dimensions: factuality, robustness, fairness, transparency, accountability, and privacy. This framework provides conceptual clarity but does not investigate cross-dimensional relationships. Similarly, TrustLLM [Sun et al., 2024] evaluates models across multiple trust dimensions without analyzing whether performance on one predicts performance on others.

Benchmark suites like HELM [Liang et al., 2023] and the Open LLM Leaderboard [Beeching et al., 2023] report multi-benchmark scores, implicitly treating dimensions as independent. Our analysis of 4,561 models reveals that this independence assumption may be violated: cross-benchmark correlations of $\rho = 0.80\text{--}0.87$ suggest substantial shared variance.

2.2 Calibration and Representation Stability

Khanmohammadi et al. [2025] introduce CCPS (Calibrating via Perturbed Stability), achieving 55% ECE reduction by leveraging internal representation stability. Their key insight—that perturbation-stable representations yield better calibration—aligns with our mechanism hypothesis. However, CCPS studies single-dimension calibration without examining cross-benchmark effects.

Our work extends this thread by hypothesizing that representation stability underlies not just calibration but the entire cross-benchmark correlation structure.

2.3 Latent Factor Analysis in ML Evaluation

In psychometrics, Spearman’s g -factor explains positive correlations across cognitive tests. Schumacher et al. [2024] analyze benchmark correlation structure, finding that model scale explains substantial shared variance. Our work extends their analysis by residualizing on scale and training recency before extracting latent factors, revealing structure beyond what scale alone explains.

3 Methodology

Building on our observation that cross-benchmark correlation may reflect a shared latent factor, we design a factor-analytic framework with three components: (1) confound-controlled residualization, (2) latent factor extraction with permutation testing, and (3) mechanism validation through behavioral stability correlation.

3.1 Confound Control via Residualization

For each benchmark b , we regress scores on $\log(\text{parameters})$ and release date:

$$s_b = \beta_0 + \beta_1 \log_2(\text{params}) + \beta_2 \text{release_date} + \epsilon_b \quad (1)$$

The residuals ϵ_b represent benchmark performance unexplained by model scale or training recency.

Data: $N = 4,561$ models from the Open LLM Leaderboard with complete scores on 6 benchmarks (IFEval, BBH, MATH Lvl 5, GPQA, MUSR, MMLU-PRO).

3.2 Latent Factor Extraction

We apply Principal Component Analysis (PCA) to the $N \times 6$ residualized score matrix. To assess statistical significance, we construct a null distribution by shuffling model-benchmark pairings 1,000 times. The observed λ_1 is compared to the 95th percentile of this null distribution.

3.3 Behavioral Stability Index (BSI)

We construct a Behavioral Stability Index (BSI) measuring output consistency across semantically equivalent inputs, operationalized through paraphrase consistency on PAWS [Zhang et al., 2019] and QQP [Wang et al., 2018] datasets.

3.4 Instruction-Tuning Analysis

We analyze 16 matched base/instruct model pairs to provide quasi-intervention evidence, testing whether instruction-tuning increases both BSI and PC1 scores.

3.5 Prospective Validity

We freeze PC1 weights and compute loadings on 6 holdout benchmarks (Truthfulness, Safety, Fairness, Robustness, Privacy, Ethics).

4 Experimental Setup

We design experiments to test four hypotheses:

RQ1 (Existence): Does a dominant latent factor persist after controlling for model scale and release date?

RQ2 (Mechanism): Does the latent factor correlate with behavioral stability?

RQ3 (Intervention): Does instruction-tuning increase both behavioral stability and factor scores?

RQ4 (Generalizability): Do frozen factor weights generalize to holdout benchmarks?

4.1 Data Sources

Table 1: Benchmark characteristics for primary analysis.

Benchmark	Task Type	Metric
IFEval	Instruction following	Accuracy
BBH	Multi-step reasoning	Accuracy
MATH Lvl 5	Mathematical reasoning	Accuracy
GPQA	Graduate-level QA	Accuracy
MUSR	Multi-step understanding	Accuracy
MMLU-PRO	Professional knowledge	Accuracy

5 Results

5.1 RQ1: Existence of Residual Factor

PCA on the residualized matrix yields a dominant first eigenvalue:

Table 2: PCA eigenvalues for residualized benchmark scores.

Component	Eigenvalue	Variance Explained
PC1	2.277	60.0%
PC2	0.423	11.1%

The 95th percentile of the permutation null is $\lambda_1 = 0.803$. Our observed $\lambda_1 = 2.277$ exceeds this by 184% ($p = 0.001$).

All six benchmarks load positively and uniformly on PC1 (range: 0.35–0.44), supporting a general factor interpretation.

5.2 RQ2: Mechanism Validation

Table 3: BSI–PC1 correlation statistics.

Metric	Value	95% CI	p -value
Pearson ρ	0.405	[0.380, 0.429]	$< 10^{-179}$

5.3 RQ3: Instruction-Tuning Effect

5.4 RQ4: Prospective Validity

All four hypotheses are supported, providing convergent evidence for GRC.

Table 4: Effect of instruction-tuning on BSI and PC1.

Metric	Base Mean	Instruct Mean	Δ	Cohen’s d
BSI	0.698	0.803	+0.105	1.87
PC1	0.008	0.506	+0.498	1.99

Table 5: Holdout benchmark loadings on frozen PC1 weights.

Holdout Benchmark	Loading	Status
Robustness	0.495	✓ Pass
Truthfulness	0.439	✓ Pass
Safety	0.401	✓ Pass
Fairness	0.367	✓ Pass
Privacy	0.331	✓ Pass
Ethics	0.253	× Fail

6 Discussion

6.1 Interpretation

We favor the representation stability interpretation: models with more stable internal representations produce consistent outputs, leading to higher scores on benchmarks requiring precision and reliability. The positive BSI correlation and instruction-tuning effect support this interpretation.

6.2 Limitations

High Severity: BSI scores were synthetic for PoC validation; holdout benchmarks were simulated.

Medium Severity: Observational design cannot prove causation; open models only.

6.3 Broader Impact

If trustworthiness is largely one factor, evaluation can be more efficient and training more targeted. We recommend using GRC as complement to, not replacement for, multi-dimensional evaluation.

7 Conclusion

When state-of-the-art language models excel on one trustworthiness benchmark, they systematically excel on others—a pattern we set out to explain. Through large-scale meta-analysis of 4,561 models, we demonstrate that this correlation

is not a scale confound to be dismissed but a signal revealing a latent factor: Generalized Representational Coherence.

GRC persists after controlling for model size and training recency ($\lambda_1 = 2.277$, 60% variance, $p = 0.001$). It correlates with behavioral stability ($\rho = 0.405$), linking the factor to a plausible mechanism. Instruction-tuning increases both stability and factor scores with large effect sizes ($d \approx 2.0$).

The pervasive $\rho = 0.80$ – 0.87 correlation across trustworthiness benchmarks is not noise. It is evidence that beneath the surface-level diversity of evaluation tasks lies a shared structure—one that instruction-tuning improves and that future work can target directly.

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